Procedure/Program:

Part 1: Identity-Based Access Using PKI

Step 1: Install Required Software

1. Update and Upgrade Ubuntu

sudo apt update && sudo apt upgrade -y

2. Install OpenSSL for Certificate Management

sudo apt install openssl -y

openssl version # Verify installation

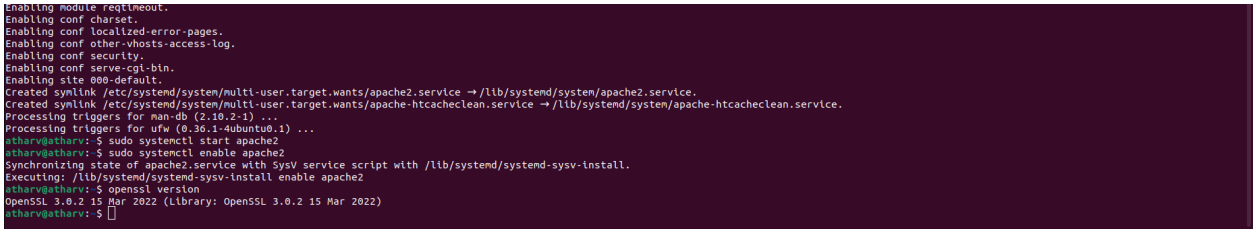
3. Install Apache Web Server

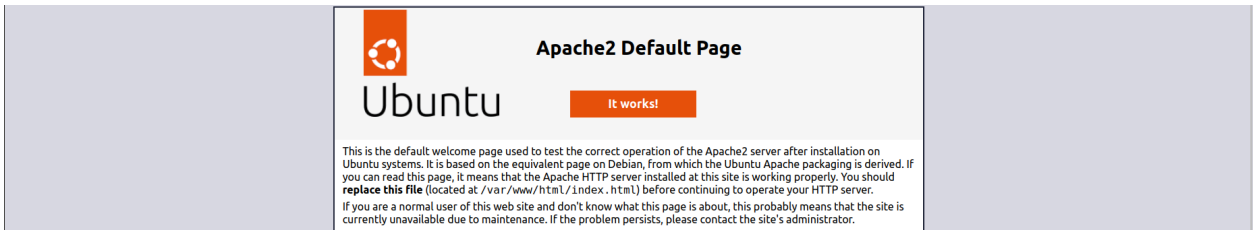
sudo apt install apache2 -y

sudo systemctl start apache2

sudo systemctl enable apache2

Test Apache: Open http://localhost in your browser. You should see the Apache default page.





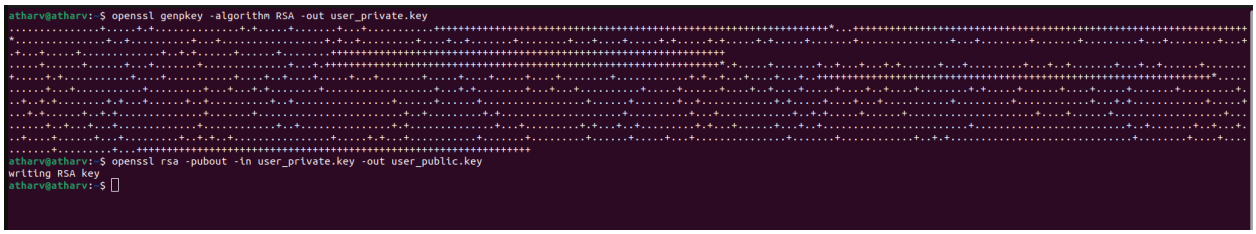
Step 2: Generate Public/Private Key Pair

1. Generate the User's Private Key

openssl genpkey -algorithm RSA -out user_private.key

2. Extract the Public Key

openssl rsa -pubout -in user_private.key -out user_public.key



Step 3: Create & Sign a User Certificate

1. Create a Certificate Signing Request (CSR)

openssl req -new -key user_private.key -out user.csr

When prompted, enter details like:

Country Name: US
Organization Name: SecureLab
Common Name: User1

2. Create a Self-Signed Certificate

openssl x509 -req -in user.csr -signkey user_private.key -out user.crt

```
atharv@atharv: $ openssl req -new -key user_private.key -out user.csr
You are about to be asked to enter information that will be incorporated
into your certificate request.
What you are about to enter is what is called a Distinguished Name or a DN.
There are quite a few fields but you can leave some blank
For some fields there will be a default value,
If you enter '.', the field will be left blank.
-----
Country Name (2 letter code) [AU]:US
State or Province Name (full name) [Some-State]:Calfornia
Locality Name (eg, city) []:LA
Organization Name (eg, company) [Internet Widgits Pty Ltd]:SecureLab
Organizational Unit Name (eg, section) []:.
Common Name (e.g. server FQDN or YOUR name) []:User1
Email Address []:xyz@gmail.com

Please enter the following 'extra' attributes
to be sent with your certificate request
A challenge password []:123456789
An optional company name []:xyz
atharv@atharv: $ openssl x509 -req -in user.csr -signkey user_private.key -out user.crt
Certificate request self-signature ok
subject=C = US, ST = Calfornia, L = LA, O = SecureLab, CN = User1, emailAddress = xyz@gmail.com
atharv@atharv: $
```

Step 4: Set Up a Certificate Authority (CA)

1. Create the CA Private Key

openssl genpkey -algorithm RSA -out ca_private.key

2. Create the CA Certificate

openssl req -key ca_private.key -new -x509 -out ca_certificate.crt

3. Sign the User's Certificate Using the CA

openssl x509 -req -in user.csr -CA ca_certificate.crt -CAkey ca_private.key -CAcreateserial -out user_signed.crt

Now, user_signed.crt is issued and signed by the CA.

```
atharv@atharv:~$ openssl genpkey -algorithm RSA -out ca_private.key
.+.+...+.........+...+..+.........+.....+......+....+...++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++*....++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++*....+......
...+........+..+...+.........+......+.....+......+..+..+.+................+............+.+............+..+.......+.....+.........+.+...+...+........+.....................+...........+.....+......+....+..
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+...++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++
atharv@atharv:~$ openssl req -key ca_private.key -new -x509 -out ca_certificate.crt
You are about to be asked to enter information that will be incorporated
into your certificate request.
What you are about to enter is what is called a Distinguished Name or a DN.
There are quite a few fields but you can leave some blank
For some fields there will be a default value,
If you enter '.', the field will be left blank.
-----
Country Name (2 letter code) [AU]:US
State or Province Name (full name) [Some-State]:california
Locality Name (eg, city) []:LA
Organization Name (eg, company) [Internet Widgits Pty Ltd]:SecureLab
Organizational Unit Name (eg, section) []:.
Common Name (e.g. server FQDN or YOUR name) []:User1
Email Address []:xyz@gmail.com
atharv@atharv:~$ openssl x509 -req -in user.csr -CA ca_certificate.crt -CAkey ca_private.key
-CAcreateseserial -out user_signed.crt
Certificate request self-signature ok
subject=C = US, ST = Calfornia, L = LA, O = SecureLab, CN = User1, emailAddress = xyz@gmail.com
-----BEGIN CERTIFICATE-----
MIIDaDCCAlACFB1RyNQVnJwStVnybcdDCZZOpn/9MA0GCSqGSIb3DQEBCwUAMHEx
CzAJBgNVBAYTAlVTMRMwEQYDVQQIDApjYWxpZm9ybmlhMQswCQYDVQQHDAJMQTES
MBAGA1UECgwJU2VjdXJlTGFiMQ4wDAYDVQQDDAVVc2VyMTEcMBoGCSqGSIb3DQEJ
ARYNeHl6QGdtYWlsLmNvbTAeFw0yNTAzMjUxNTI0MDBaFw0yNTA0MjQxNTI0MDBa
MHAxCzAJBgNVBAYTAlVTMRIwEAYDVQQIDAlDYWxmb3JuaWExCzAJBgNVBAcMAkxB
MRIwEAYDVQQKDAlTZWN1cmVMYWIxDjAMBgNVBAMMBVVzZXIxMRwwGgYJKoZIhvcN
AQkBFg14eXpAZ21haWwuY29tMIIBIjANBgkqhkiG9w0BAQEFAAOCAQ8AMIIBCgKC
AQEAw7pMdiuPHiwFChs+EGanittmpi3FNfsY6TdvzT3FdbTVguK4s7CvHzqyFt+X
9B/k/DyZTrmap9Ws6NUpVPHf1UQ0rPqkXzs3vqkIfjxEPIcdsrW0TOIK6QC/fsRE
uGgTlG8QHkx3mo93A+wQue5sOUdVrzovDo8y8bwKuyEm3tOwlkssQ/lD5pyHzsOF
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6BUANMKUilIRhASAFsF44Ba7RUTUYHKazMUSyNIov4Clk1b8gO6ckHU/cpI6kbnK
w2xVinf39wOnWGNdicU5kLG7wQIDAQABMA0GCSqGSIb3DQEBCwUAA4IBAQCl0FDq
rsUTd7NfR3IvhG7HWIXNebAv5oR5QojNmD/nOhVpBPuB3OvPiHplxW8nzBnCB7wg
ljVNOqZNAosy9X94/F1a7C3jq9rOdor8A052AfMgqMxlJn7qI1z7A5Vq6N9wHI4l
WlMrVnct3XI4pb9zkAU04w7E49hQtVYl6rvvUFA59z2lcpVRs6nnesgrmrLh3++j
EGxz1UXfVfIs2jKmITkXO5Ce/WozSajuC0lleccQUd3HBvfu1FCF2aj9NRjJeCf0
wz1ILAnOPKjKKXZY3Cp4Y8AAuI10BTUPTaF1wsOekJlmEhMerpNh3SG6yTiohr/B
dVRUA/MyCWGRwr0k
-----END CERTIFICATE-----
-CAcreateserial: command not found
atharv@atharv:~$
```

Step 5: Configure Apache for Identity-Based Access

1. Enable SSL Module

sudo a2enmod ssl

2. Configure SSL Certificate in Apache

Edit the Apache SSL configuration:

sudo nano /etc/apache2/sites-available/default-ssl.conf

Replace <VirtualHost *:443> section with:

<VirtualHost *:443>
DocumentRoot /var/www/html
SSLEngine on
SSLCertificateFile /etc/ssl/certs/user_signed.crt
SSLCertificateKeyFile /etc/ssl/private/user_private.key
SSLCACertificateFile /etc/ssl/certs/ca_certificate.crt
SSLVerifyClient require
SSLVerifyDepth 1
</VirtualHost>

3. Restart Apache

sudo systemctl restart apache2

```
atharv@atharv:~$ sudo a2enmod ssl
Considering dependency setenvif for ssl:
Module setenvif already enabled
Considering dependency mime for ssl:
Module mime already enabled
Considering dependency socache_shmcb for ssl:
Enabling module socache_shmcb.
Enabling module ssl.
See /usr/share/doc/apache2/README.Debian.gz on how to configure SSL and create self-signed certificates.
To activate the new configuration, you need to run:
  systemctl restart apache2
atharv@atharv:~$ sudo nano /etc/apache2/sites-available/default-ssl.conf
atharv@atharv:~$ sudo systemctl restart apache2
atharv@atharv:~$
```

Step 6: Create a Protected Directory

1. Create a Secure Folder

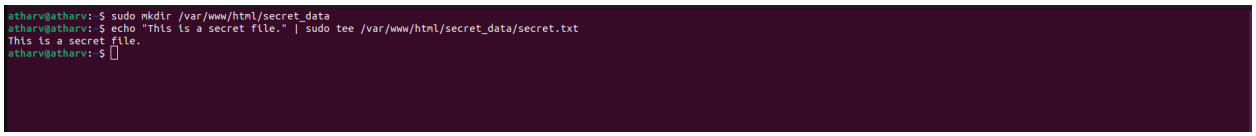
sudo mkdir /var/www/html/secret_data

echo "This is a secret file." | sudo tee /var/www/html/secret_data/secret.txt

2. Restrict Access to the Directory

Apache will require valid certificates to access /secret_data/.

Apache is now configured for IBAC!



Step 7: Simulate the Confinement Problem

1. Create Python Script

nano access_control.py

2. Add the Following Code

```
# Function to check access based on clearance levels
def check_access(user_clearance, resource_clearance):
if user_clearance >= resource_clearance:
return "Access Granted"
Else:
return "Access Denied"
# Clearance Levels
clearance_levels = {
1: "Top Secret",
2: "Secret",
3: "Confidential",
4: "Unclassified"
}
# Get user input
print("Available Clearance Levels:")
for level, name in clearance_levels.items():
print(f"{level}: {name}")
user_clearance = int(input("Enter your clearance level (1-4): "))
resource_clearance = int(input("Enter the clearance level of the resource (1-4):
"))
# Check Access
access_result = check_access(user_clearance, resource_clearance)
# Display Result
print(f"Your Clearance Level: {clearance_levels[user_clearance]}")
print(f"Resource Clearance Level: {clearance_levels[resource_clearance]}")
20print(access_result)
```

3. Code Run the Script
python3 access_control.py

```
  GNU nano 6.2                                                    access_control.py *
# Function to check access based on clearance levels
def check_access(user_clearance, resource_clearance):
    if user_clearance >= resource_clearance:
        return "Access Granted"
    else:
        return "Access Denied"

# Clearance Levels
clearance_levels = {
    1: "Top Secret",
    2: "Secret",
    3: "Confidential",
    4: "Unclassified"
}

# Get user input
print("Available Clearance Levels:")
for level, name in clearance_levels.items():
    print(f"{level}: {name}")

user_clearance = int(input("Enter your clearance level (1-4): "))
resource_clearance = int(input("Enter the clearance level of the resource (1-4): "))

# Check Access
access_result = check_access(user_clearance, resource_clearance)

# Display Result
print(f"Your Clearance Level: {clearance_levels[user_clearance]}")
print(f"Resource Clearance Level: {clearance_levels[resource_clearance]}")
print(access_result)

File Name to Write: access_control.py
```

```
atharv@atharv:~$ nano access_control.py
atharv@atharv:~$ python3 access_control.py
Available Clearance Levels:
1: Top Secret
2: Secret
3: Confidential
4: Unclassified
Enter your clearance level (1-4): 1
Enter the clearance level of the resource (1-4): 1
Your Clearance Level: Top Secret
Resource Clearance Level: Top Secret
Access Granted
atharv@atharv:~$ python3 access_control.py
Available Clearance Levels:
1: Top Secret
2: Secret
3: Confidential
4: Unclassified
Enter your clearance level (1-4): 2
Enter the clearance level of the resource (1-4): 3
Your Clearance Level: Secret
Resource Clearance Level: Confidential
Access Denied
atharv@atharv:~$ python3 access_control.py
Available Clearance Levels:
1: Top Secret
2: Secret
3: Confidential
4: Unclassified
Enter your clearance level (1-4): 4
Enter the clearance level of the resource (1-4): 1
Your Clearance Level: Unclassified
Resource Clearance Level: Top Secret
Access Granted
atharv@atharv:~$
```